Fast approximation of Frechet distance for imbalanced curves

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Editorial scope. A query-dependent greedy simplification has at most twice the short curve complexity and can be computed in linear time; its full matching argument yields a three-gap decision procedure, which the cited optimization conversion turns into the main running-time guarantee. Official CCBY mathematical material only; complete selected main proofs retained. Omitted-proof oracle/Section4 route not counted or used.

Allowed standard prerequisites (editorial).

- Orthogonal vectors hypothesis, standard Frechet distance/matching facts and approximate optimization conversion. Original official definitions and references retained. Full source-specific reductions, gadgets and simplification/matching proof included. Unproved oracleTheorem11 and Section4 simplificationLemma12 are excluded and not relied on.

1 Introduction

The Fréchet distance is a distance metric for polygonal curves, where each curve is represented as a piecewise-linear interpolation between its vertices. There are different variants of this metric and the two most common variants are the discrete and the continuous Fréchet distance. Alt and Godau [3] were the first to study the computational complexity of the Fréchet distance. Let n and m denote the number of vertices of the two curves, which we refer to as their complexities, with $n \geq m$. They showed that the continuous Fréchet distance can be computed in $\mathcal{O}(nm \log n)$ time. Eiter and Mannilla [21] defined the discrete Fréchet distance and gave an $\mathcal{O}(nm)$ time algorithm. Although both results were slightly improved later [15, 11, 1], Bringmann [6] showed that for d-dimensional curves with $d \geq 2$ it is very unlikely that significantly faster algorithms exists: unless the Strong Exponential Time Hypothesis (SETH) fails, no (1.001)-approximation algorithm for the continuous or discrete Fréchet distance can run in $\mathcal{O}((nm)^{1-\delta})$ time for any $\delta > 0$. Later, Bringmann and Mulzer [10] extended this result for the discrete Fréchet distance in 1D and increased the approximation factor up to $1.4 - \varepsilon$ for any

$\varepsilon > 0$111In [10], this result is stated only for $n = m$, but the same construction works for the case $n \neq m$ as well.. For the continuous and the discrete Fréchet distance in 1D, Buchin, Ophelders, and Speckmann [12] showed a hardness result for approximation factors up to $3 - \varepsilon$, but only in the case where $n = m$. A recent result by Blank and Driemel [5] shows that the continuous Fréchet distance in 1D can indeed be computed faster when $m = n^\alpha$ with $\alpha \in (0, 1)$, namely in $\mathcal{O}(n \log n + m^2 \log^2 n)$ time.

Related Work.

In recent years, many works have studied approximation algorithms for the Fréchet distance [18, 26, 27]. This line of research culminated in the first constant-factor approximation algorithm for the Fréchet distance that runs in subquadratic time, due to Cheng, Huang, and Zhang [17]. Their randomized $(7+\varepsilon)$-approximation algorithm for the continuous Fréchet distance runs in $\mathcal{O}(nm^{0.99} \log(n/\varepsilon))$ time for any $\varepsilon > 0$ and succeeds with probability at least $1 - 1/n^6$.

A simple idea to 3-approximate the Fréchet distance between curves where $m = n^\alpha$ with $\alpha \in (0, 1)$ in subquadratic time is the following. Compute a curve P' of complexity $n' \in o(n)$ and its distance δ to P such that every curve Q of complexity m has Fréchet distance at least δ to P. Then compute the Fréchet distance δ' between P' and Q. Using triangle inequality, one can show that the Fréchet distance between P and Q is at least $\max\{\delta'/2, \delta\}$ and at most $3 \cdot \max\{\delta'/2, \delta\}$, which yields a 3-approximation. The difficulty of this approach is to compute such a curve P'. Two related variants of this simplification problem have been studied. The first is computing an optimal m-simplification of P, that is, a curve P^* of complexity at most m minimizing the Fréchet distance to P among all curves of complexity at most m. The second is computing an optimal δ-simplification, that is, a curve P^* of minimum complexity among all curves that have Fréchet distance at most δ to P. However, no previous result achieved a subquadratic running time for curves in arbitrary dimension when the approximation factor on the complexity of the simplified curve is constant [2, 4, 7, 13, 16, 22, 28].

A different approach to reduce the running time of computing the Fréchet distance is via Fréchet distance oracles. Such an oracle is a data structure that preprocesses an input curve and subsequently answers Fréchet distance queries with respect to query curves. Driemel and Har-Peled [19] described a data structure of size $\mathcal{O}(n \log n)$ using $\mathcal{O}(m^2 \log n \log(m \log n))$ query time, which approximates the continuous Fréchet distance up to a (large) constant factor. For exact queries, [14] presented a data structure of size $\mathcal{O}(nm)^{\mathsf{poly}(d,m)}$ and query time in $\mathcal{O}((md)^{\mathcal{O}(1)} \log(nm))$, where the query complexity m is given at preprocessing time. Blank and Driemel [5] showed that for curves in 1D, there is a data structure of size $\mathcal{O}(n \log n)$ that computes the continuous Fréchet distance exactly in $\mathcal{O}(m^2 \log^2 n)$ time. For the discrete Fréchet distance, Filtser and Filtser [22] improved [20] and presented a data structure with size in $\mathcal{O}(1/\varepsilon)^{kd} \log(\varepsilon)^{-1}$ that approximates the discrete Fréchet distance

within a factor of $(1+\varepsilon)$ using expected $\widetilde{\mathcal{O}}(kd)$ query time, where $\widetilde{\mathcal{O}}(\cdot)$ hides polylogarithmic factors in n. Gudmundsson, Seybold, and Wong [24] used the lower bound construction by Bringmann [6] to show in a similar fashion to [8, 20, 25] the following. In 2D, there exists no Fréchet distance oracle with poly(n) preprocessing time and $\mathcal{O}((nm)^{1-\delta})$ query time and approximation factor 1.001 unless SETH fails.

Table 1: Upper and Lower Bounds for the Fréchet distance Computation. Upper Bounds
| Discrete | Continuous
1D | $\mathcal{O}(mn)$ | exact | [21, 1] | $\widetilde{\mathcal{O}}(mn)$ | exact | [3, 11, 15]
$\widetilde{\mathcal{O}}(n+m^2)$ | 2-approx. | Cor.20 | $\widetilde{\mathcal{O}}(n+m^2)$ | exact | [5]
dD | $\mathcal{O}(mn)$ | exact | [21, 1] | $\widetilde{\mathcal{O}}(mn)$ | exact | [3, 11, 15]
$\widetilde{\mathcal{O}}(n+m^2)$ | $(3+\varepsilon)$-approx. | Thm.23 | $\widetilde{\mathcal{O}}(n+m^2)$ | $(3+\varepsilon)$-approx. | Thm.23
$\widetilde{\mathcal{O}}(nm^{0.99})$ | $(7+\varepsilon)$-approx. | [17] | $\widetilde{\mathcal{O}}(nm^{0.99})$ | $(7+\varepsilon)$-approx. | [17]
Lower Bounds
| Discrete | Continuous
1D | $\nexists\, \mathcal{O}(m^{2-\delta})$ | $(3-\varepsilon)$-approx. | [12] | $\nexists\, \mathcal{O}(m^{2-\delta})$ | $(3-\varepsilon)$-approx. | [12]
$\nexists\, \widetilde{\mathcal{O}}((mn)^{1-\delta})$ | $(1.4-\varepsilon)$-approx. | [10] | | |
$\nexists\, \widetilde{\mathcal{O}}((mn)^{1-\delta})$ | $(2-\varepsilon)$-approx. | Thm.9 | | |
2D | $\nexists\, \mathcal{O}((mn)^{1-\delta})$ (1.001)-approx. [6]
$\nexists\, \mathcal{O}((mn)^{1-\delta})$ $(1+\sqrt{2}-\varepsilon)$-approx. in L_2 Thm.10
$\nexists\, \mathcal{O}((mn)^{1-\delta})$ $(3-\varepsilon)$-approx. in L_∞ Thm.10

Our Results.

Our first set of results shows lower bounds for algorithms that approximate the Fréchet distance conditioned on the Orthogonal Vectors Hypothesis (OVH). We show that the discrete Fréchet distance between 1-dimensional curves cannot be $(2-\varepsilon)$-approximated in $\mathcal{O}((nm)^{1-\delta})$ time for any $\varepsilon, \delta > 0$ unless OVH fails. This increases the approximation factor of $1.4-\varepsilon$ by Bringmann and Mulzer [10]. Using a similar construction in 2D, we show that the continuous and discrete Fréchet distance cannot be approximated within a factor less than $1+\sqrt{2}$ (resp. 3) when the curves lie in the Euclidean space (resp. in the L_∞-space) in $\mathcal{O}((nm)^{1-\delta})$. This leads to lower bounds for Fréchet distance oracles as well and answers an open question by [24].

To obtain this first set of results, we reduce from the Orthogonal Vector Problem: Given two sets of vectors $U, V \subset \{0,1\}^d$ of size m and n, do there exist two vectors $u \in U$ and $v \in V$ such that $\langle u, v\rangle = 0$? Then, the results follow by OVH, which is implied by SETH [29, 9].

Definition 1 (Orthogonal Vectors Hypothesis (OVH)).

For all $\delta > 0$ there exists a $c > 0$ such that there is no algorithm solving OV instances $U, V \subset \{0,1\}^d$ of size m and n with $m \leq n$ and $d = c \log n$ in time $\mathcal{O}((nm)^{1-\delta})$.

We construct a curve P of complexity $\mathcal{O}(n+m)$ depending on V and the size of U and a curve Q of complexity $\mathcal{O}(m)$ depending only on U. Then, we show that the Fréchet distance between P and Q is

at most 1 if and only if the OV instance has a solution. In order to get a lower bound also for the case where $n \gg m$, we construct P such that the main part of P lies in a ball of radius 1. Hence, this main part can be matched to a single point of Q. Further, to obtain an inapproximability result of $2-\varepsilon$ in 1D, the points of P and Q have integer coordinates. Hence, there are only three different values that the vertices of the main part of P can have. This significantly restricts our possibilities to define P and is the main difference to the construction by Bringmann and Mulzer [10]. They also note in their paper that obtaining a lower bound for an approximation factor of 1.4 or higher requires a significantly different construction compared to theirs.

On the positive side, we present a data structure that preprocesses a 1-dimensional curve of complexity n in $\mathcal{O}(n \log n)$ time using $\mathcal{O}(m)$ space that 2-approximates the discrete Fréchet distance to query curves of complexity m in $\mathcal{O}(m^2 \log m)$ time. Hence, there exists a 2-approximation algorithm for the discrete Fréchet distance in $\mathcal{O}(n \log n + m^2 \log m)$ time matching our approximation factor lower bound and essentially the time lower bound by [12]. Secondly, we match our approximation factor lower bound in the L_∞-space by giving a simple $(3+\varepsilon)$-approximation algorithm for the discrete and continuous Fréchet distance using $\mathcal{O}((n+m^2) \log n)$ time for curves in any dimension under any L_p-norm. Table 1 summarizes the results of this paper and related work.

2 Preliminaries

For any two points $p_1, p_2 \in \mathbb{R}^d$, $\overline{p_1 p_2}$ denotes the directed line segment connecting p_1 with p_2. A polygonal curve P of complexity n is formed by ordered line segments $\overline{P(i)P(i+1)}$ of points $P(1), P(2), \ldots, P(n)$, where $P(i) \in \mathbb{R}^d$. It can be viewed as a function $P : [1, n] \to \mathbb{R}^d$, where $P(i+\alpha) = (1-\alpha)P(i) + \alpha P(i+1)$ for $i \in \{1, \ldots, n\}$ and $\alpha \in [0, 1]$. This curve is also denoted with $\langle P(1), \ldots, P(n) \rangle$. We call the points $P(i)$ vertices and the line segments $\overline{P(i)P(i+1)}$ edges of P. Further, $P[s, t]$ is the subcurve of P obtained from restricting the domain to the interval $[s, t]$. Two curves $P = \langle P(1), \ldots, P(n) \rangle$ and $Q = \langle Q(1), \ldots, Q(m) \rangle$ can be composed into the curve $P \circ Q = \langle P(1), \ldots, P(n), Q(1), \ldots, Q(m) \rangle$. For a number $k \in \mathbb{N}_0$, define P^k be the composition of k copies of P.

Let $P : [1, n] \to \mathbb{R}^d$ and $Q : [1, m] \to \mathbb{R}^d$ be two polygonal curves. A continuous matching M between P and Q is defined by two functions $h_P \in \mathcal{F}_n$ and $h_Q \in \mathcal{F}_m$ and consists of the tuples $(h_P(a), h_Q(a))$ for $a \in [0, 1]$, where $\mathcal{F}_n$ is the set of all continuous, non-decreasing functions $h : [0, 1] \to [1, n]$ with $h(0) = 1$ and $h(1) = n$, respectively $\mathcal{F}_m$ for m. Then, the (continuous) Fréchet distance between P and Q is defined as

$$d_{\mathrm{F}}(P, Q) = \min_{\text{continuous matching } M} \max_{(x,y) \in M} \|P(x) - Q(y)\|.$$

For a fixed δ, the free space $F_\delta(P, Q)$ is a subset of $[1, n] \times [1, m]$ such that $(x, y) \in F_\delta(P, Q)$ if and only if $\|P(x) - Q(y)\| \leq \delta$. Points in $([1, n] \times [1, m]) \setminus F_\delta(P, Q)$ are contained in the non-free space. Hence,

the continuous Fréchet distance between P and Q is at most δ if a continuous matching $M \subset F_\delta(P,Q)$ between P and Q exists.

A discrete matching M between P and Q is an ordered sequence that starts in $(1,1)$, ends in (n,m) and for any consecutive tuples (i,j), (i',j') in M it holds that $(i',j') \in \{(i+1,j),(i,j+1),(i+1,j+1)\}$. Then, the discrete Fréchet distance between P and Q is

$$d_{\mathrm{dF}}(P,Q) = \min_{\text{discrete matching } M} \max_{(i,j)\in M} \|P(i) - Q(j)\|.$$

The free space matrix M_δ is an $n \times m$ matrix, where $M_\delta[i,j] = 1$ if $|P(i) - Q(j)| \leq \delta$ and otherwise $M_\delta[i,j] = 0$. We say that (i,j) is reachable if $d_{\mathrm{dF}}(P[1,i], Q[1,j]) \leq \delta$.

Observation 2.
Let P_1, P_2, Q_1, Q_2 be curves. If $d_{\mathrm{F}}(P_1,Q_1) \leq \delta$ and $d_{\mathrm{F}}(P_2,Q_2) \leq \delta$, then $d_{\mathrm{F}}(P_1 \circ P_2, Q_1 \circ Q_2) \leq \delta$. For the discrete Fréchet distance, the analogue statement holds.

Observation 3.
Let P and Q be polygonal curves, then $d_{\mathrm{F}}(P,Q) \leq d_{\mathrm{dF}}(P,Q)$.

3 Approximation lower bounds for the imbalanced case

In this section, we show a lower bound for the discrete Fréchet distance in 1D and for the discrete and continuous Fréchet distance in 2D depending on OVH. The construction for both bounds is very similar and we start with giving a schematic view of the reduction.

Let $U, V \subset \{0,1\}^d$ be two sets of size m and n, where $m \leq n$ and denote with v_{ik} (resp. u_{ik}) the k-th bit of the i-th vector of V (resp. U). We construct two 1-dimensional (resp. 2-dimensional) curves P and Q of complexity $\mathcal{O}(nd)$ and $\mathcal{O}(md)$, where P depends on V and Q on U. We show that their discrete (and continuous) Fréchet distance is at most 1 if and only if two vectors that are orthogonal exist in V and U. Define

$$P = P_s \circ (P^*)^{m-1} \circ P_{null} \circ \bigcirc_{i=1}^{n} \left((P_0)^{d+1} \circ P_1 \circ \bigcirc_{\ell=1}^{d} P_{v_{i\ell}} \circ P_1\right) \circ P_{null} \circ (P^*)^{m-1} \circ P_s,$$

$$Q = (Q^*)^{m-1} \circ \bigcirc_{j=1}^{m} \left(Q_{null} \circ Q_c \circ (Q_1)^{d+1} \circ Q_0 \circ \bigcirc_{\ell=1}^{d} Q_{u_{j\ell}} \circ Q_0 \circ Q_c\right) \circ Q_{null} \circ (Q^*)^{m-1},$$

where the subcurves are defined later. Since the complexity of P can be arbitrarily larger than the complexity of Q, we construct the curve P such that the main part of P, i.e., in between the two subcurves P_{null}, is contained in a ball of radius 1. Then, we can match this whole part to a single point on Q. This point is Q_c in our construction. At a high level idea, we want to enforce that in a matching that realizes the Fréchet distance, the two subcurves P_{null} get roughly matched to different subcurves Q_{null}. Then, the subcurve in between the two points matched to P_{null} contains $(Q_1)^{d+1}$ as a subcurve. Now, we enforce that $(Q_1)^{d+1}$ gets roughly matched to $(P_0)^{d+1}$. Hence, afterwards $\bigcirc_{k=1}^{d} P_{v_{ik}}$ and $\bigcirc_{\ell=1}^{d} Q_{u_{j\ell}}$ have to be matched together for some i and j. We will show that this is only possible if $\langle v_i, u_j \rangle = 0$. This matching is schematically visualized in Figure 1.

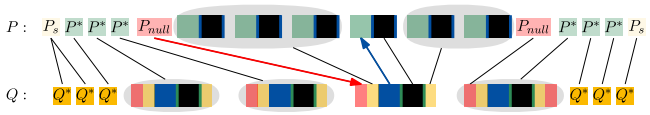


Figure 1: A schematic view of the construction of the curves P and Q. Here, Q_{null} is drawn in red, P_0 and Q_0 in green, P_1 and Q_1 in blue, and Q_c in orange. We define

$$P_s = \langle(-1,0)\rangle, P_{null} = \langle(-5,0)\rangle, \quad P^* = \langle(-3,0)\rangle \circ \langle P_0\rangle^{2d+3} \circ \langle(-3,0)\rangle,$$
$$Q_c = \langle(0,0)\rangle, Q_{null} = \langle(-4,0)\rangle, \quad Q^* = \langle(-2,0),(0,0),(-2,0)\rangle,$$

for 2D and the definition carries over directly to 1D. The curves P_0, P_1, Q_0, and Q_1 are visualized in Figure 2. For the 1D setting, let

$$P_0 = \langle -1, 1\rangle, P_1 = \langle -1, 0\rangle, \quad Q_0 = \langle -2, 1\rangle, Q_1 = \langle -2, 2\rangle.$$

For the 2D constructions, define eight points with distance 1 to $(0,0)$ where $a_1 = (-1,0)$ and the direction of a_{i+1} is the direction of a_i rotated by $\pi/4$, i.e., in L_2

$$a_1 = (-1,0), a_2 = (-1/\sqrt{2}, -1/\sqrt{2}), \quad a_3 = (0,-1), a_4 = (1/\sqrt{2}, -1/\sqrt{2}),$$
$$a_5 = (1,0), a_6 = (1/\sqrt{2}, 1/\sqrt{2}), \quad a_7 = (0,1), a_8 = (-1/\sqrt{2}, 1/\sqrt{2}),$$

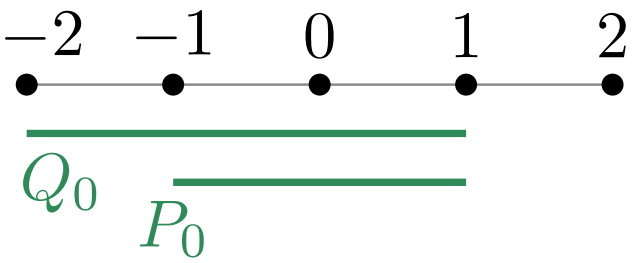
and in L_∞

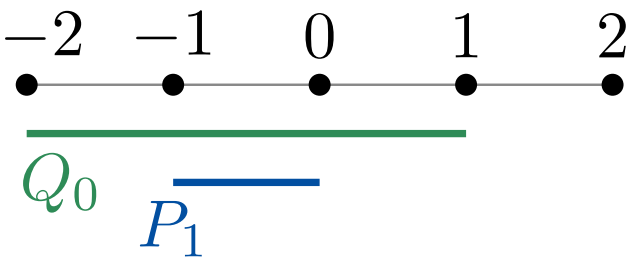
$$a_1 = (-1,0), a_2 = (-1,-1), \quad a_3 = (0,-1), a_4 = (1,-1),$$
$$a_5 = (1,0), a_6 = (1,1), \quad a_7 = (0,1), a_8 = (-1,1).$$

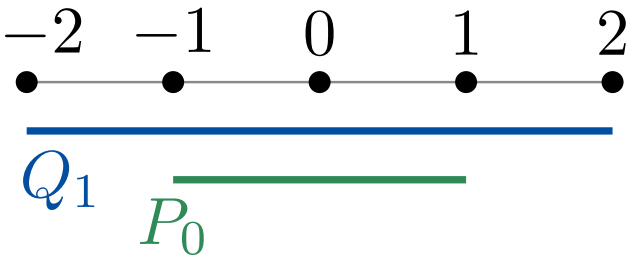
Define $c = \sqrt{2}$ and $\tilde{a} = a_8 + a_7$ in L_2 and $c = 2$ and $\tilde{a} = 2a_8$ in L_∞. Note that c is defined such that ca_1 is the point with distance 1 to a_8 and a_2 that maximizes the distance to $(0,0)$ and $\tilde{a}$ such that $\|\tilde{a} - a_2\| = 1 + c$ and $\|\tilde{a} - a_8\| = 1$. Then we set

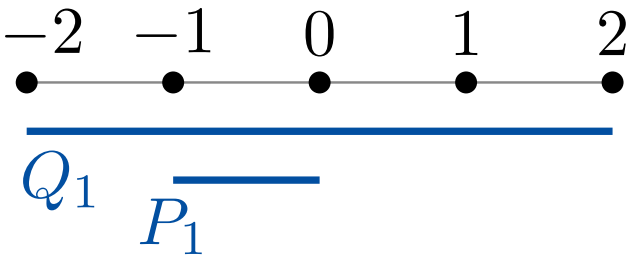
$$P_0 = \langle a_1, a_2, \ldots a_7, a_8, a_7 \ldots, a_1\rangle, P_1 = \langle a_1, a_2, \ldots, a_8, a_1, a_2, a_1 a_8, \ldots, a_1\rangle,$$
$$Q_0 = \langle 2a_1, 2a_2, \ldots, 2a_8, ca_1, 2a_8, \ldots, 2a_1\rangle, Q_1 = \langle 2a_1, 2a_2, \ldots, 2a_7, \tilde{a}, 2a_7, \ldots, 2a_1\rangle.$$

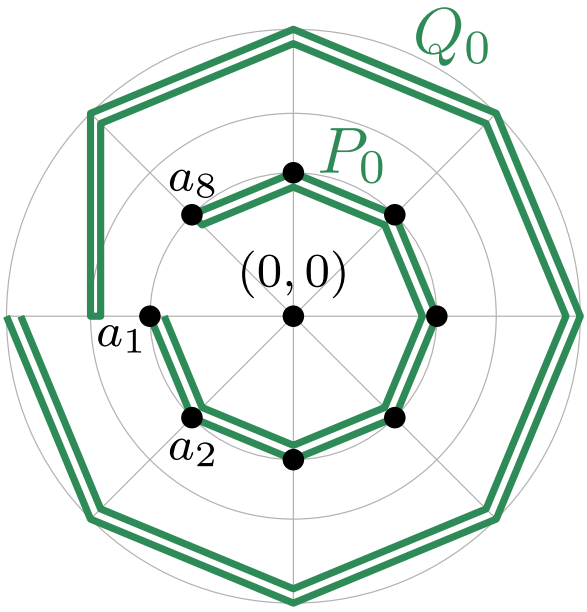
We show that this results in an inapproximability result of $(1 + c - \varepsilon)$ for time in $\mathcal{O}((nm)^{1-\delta})$.

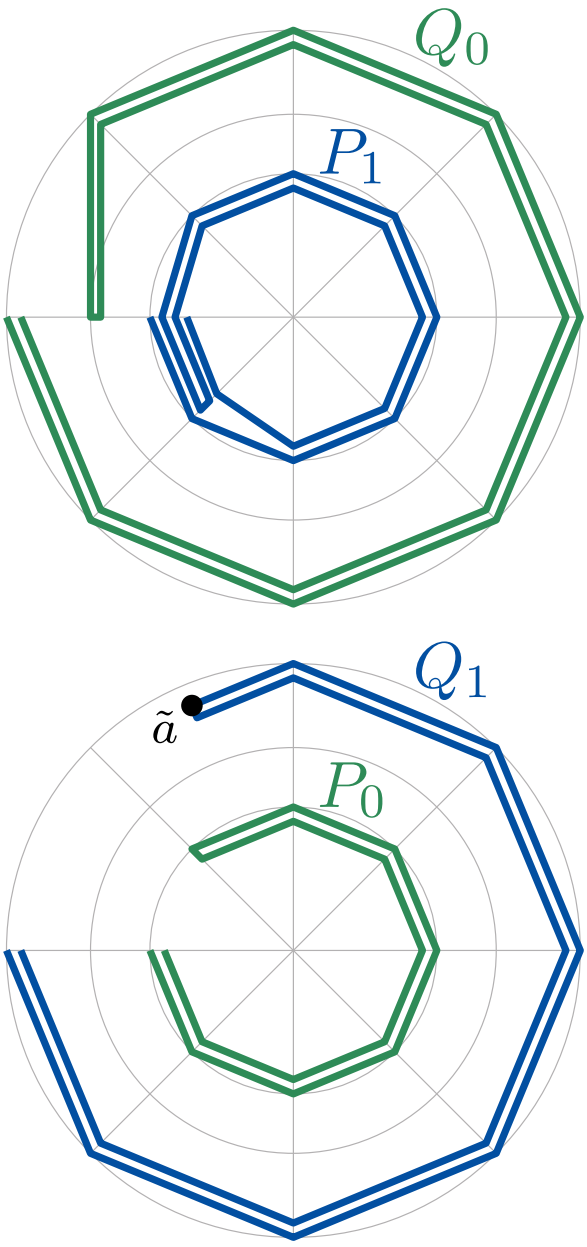


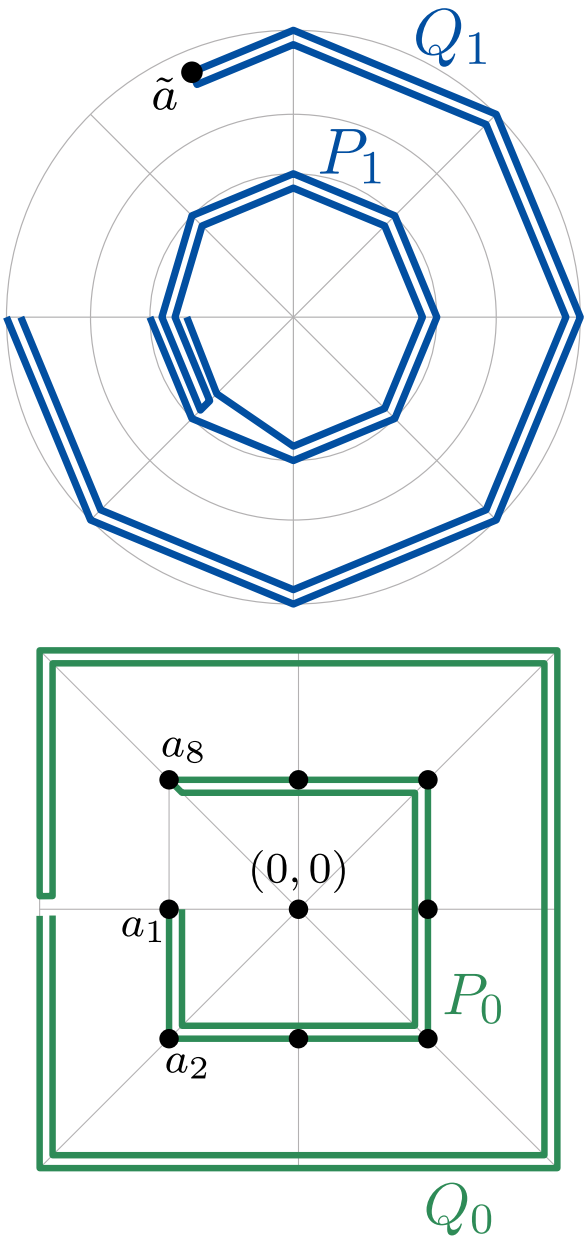

−2
−1
0
1
2
Q_0
P_1

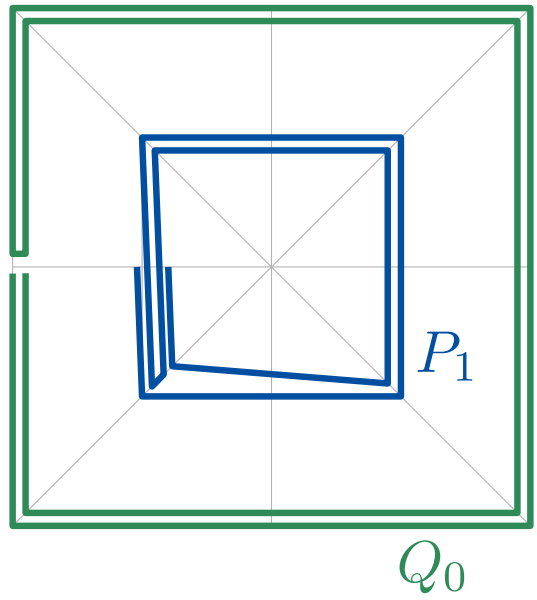

−2
−1
0
1
2
Q_1
P_0

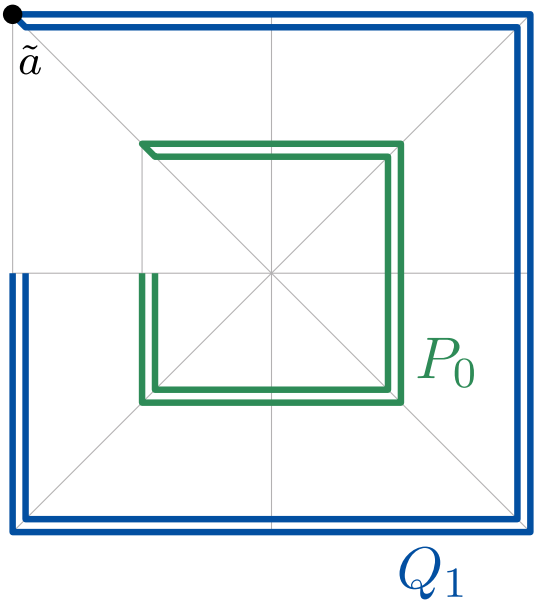

−2
−1
0
1
2
Q_1
P_1


Q_0
P_0
a_8
$(0,0)$
a_1
a_2


Q_0
P_1
Q_1
$\tilde{a}$
P_0


Q_1
$\tilde{a}$
P_1
a_8
$(0,0)$
a_1
P_0
a_2
Q_0


P_1
Q_0


$\tilde{a}$
P_0
Q_1

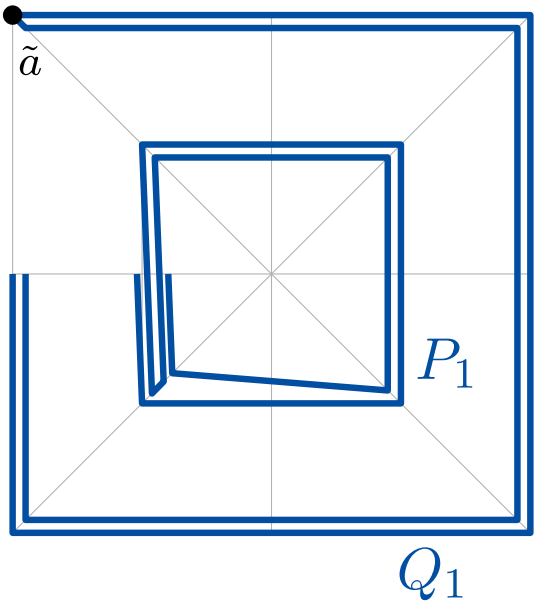


Figure 2: The curves P_0, P_1, Q_0, and Q_1. The upper curves are defined in 1D, the middle curves for the L_2-metric in 2D and the lower curves for the L_∞-metric in 2D.

Lemma 4.

If there exist indices i and j such that $\langle v_i, u_j \rangle = 0$, then in every construction for P and Q above it holds that $d_{\mathrm{dF}}(P, Q) \leq 1$.

Proof.

For every construction of P and Q, it holds that $d_{\mathrm{dF}}(P^*, Q^*) \leq 1$, $d_{\mathrm{dF}}(P_s, Q^*) \leq 1$. Further, for $x, y \in \{0, 1\}$ with $\langle x, y \rangle = 0$, it holds that $d_{\mathrm{dF}}(P_x, Q_y) \leq 1$. Then, we can match the following with Fréchet distance at most 1 and the lemma follows by Observation 2

$$
\begin{aligned}
&\blacksquare P_s \circ (P^*)^{m-j} && (Q\\
&\blacksquare (P^*)^{j-1} \circ P_{null} && \textstyle\bigcirc_{k=1}^{j-1}(Q_{null} \circ Q_c \circ (Q_1)^{d+1}\\
&\textstyle\bigcirc_{\ell=1}^{d} Q_{u_{k\ell}} \circ Q_0 \circ Q_c) \circ Q_{null}, && \\
&\blacksquare \textstyle\bigcirc_{k=1}^{i-1} \left((P_0)^{d+1} \circ P_1 \circ \bigcirc_{\ell=1}^{d} P_{v_{k\ell}} \circ P_1\right) && \\
&\blacksquare (P_0)^{d+1} \circ P_1 \circ \textstyle\bigcirc_{\ell=1}^{d} P_{v_{i\ell}} \circ P_1 && (Q_1)^{d+1} \circ Q_0 \circ \textstyle\bigcirc_{\ell=1}^{d} Q_{u_j}\\
&\blacksquare \textstyle\bigcirc_{k=i+1}^{n} \left((P_0)^{d+1} \circ P_1 \circ \bigcirc_{\ell=1}^{d} P_{v_{k\ell}} \circ P_1\right) && \\
&\blacksquare P_{null} \circ (P^*)^{m-j} && \textstyle\bigcirc_{k=j+1}^{m}(Q_{null} \circ Q_c \circ (Q_1)^{d+1}\\
&\textstyle\bigcirc_{\ell=1}^{d} Q_{u_{k\ell}} \circ Q_0 \circ Q_c) \circ Q_{null}, && \\
&\blacksquare (P^*)^{j-1} \circ P_s && (Q^*
\end{aligned}
$$

◀

To show the other direction, we first prove some lemmas.

Lemma 5.

Consider the 1D setting and let $Q' = \bigcirc_{\ell=1}^{d'} Q_{\sigma_\ell}$, where $\sigma_\ell \in \{0,1\}$ for every ℓ. Further, let $s \leq t$ be such that $P[s,t]$ is a subcurve in between the two subcurves P_{null} and $d_{\mathrm{dF}}(Q', P[s,t]) < 2$. Then $P[s,t] = \bigcirc_{\ell=1}^{d'} P_{\pi_\ell}$ where $\pi_\ell = 0$ if $\sigma_\ell = 1$ and $\pi_\ell \in \{0,1\}$ if $\sigma_\ell = 0$ for every ℓ.

Proof.

It holds that every second vertex of Q' is -2 and the vertices of value -1 on $P[s,t]$ are the only vertices on $P[s,t]$ within distance less than 2 to those vertices of Q'. Further, it holds that the vertices with value -1 of $P[s,t]$ have distance less than two only to the vertices of value -2 of Q'. Hence, since the vertices of value -1 on $P[s,t]$ are exactly every second vertex on $P[s,t]$, we must traverse $P[s,t]$ and Q' synchronously. Therefore, if $\sigma_\ell = 0$ then $\pi_\ell \in \{0,1\}$ and if $\sigma_\ell = 1$ then $\pi_\ell = 0$ by construction. ◀

In 2D, we also want to show a bound for the continuous Fréchet distance. Then the vertices of Q can also be matched to points on edges of P. We use the notation of the following subcurve of P_0 (resp. P_1): $P_0' = \langle a_4, a_5, \ldots, a_8, \ldots, a_4 \rangle$ (resp. $P_1' = \langle a_4, \ldots, a_8, a_1, a_2, a_1, a_8, \ldots, a_4 \rangle$).

Lemma 6.

Consider the 2D setting. For any subcurve P' of P in between the two subcurves P_{null} with $d_{\mathrm{F}}(P', Q_y) < 1 + c$ for $y \in \{0,1\}$, it holds that P_x' is a subcurve of P' and P' is a subcurve of $\langle a_4, a_3, a_2 \rangle \circ P_x \circ \langle a_2, a_3, a_4 \rangle$, where $x = 0$ if $y = 1$ and $x \in \{0,1\}$ if $y = 0$.

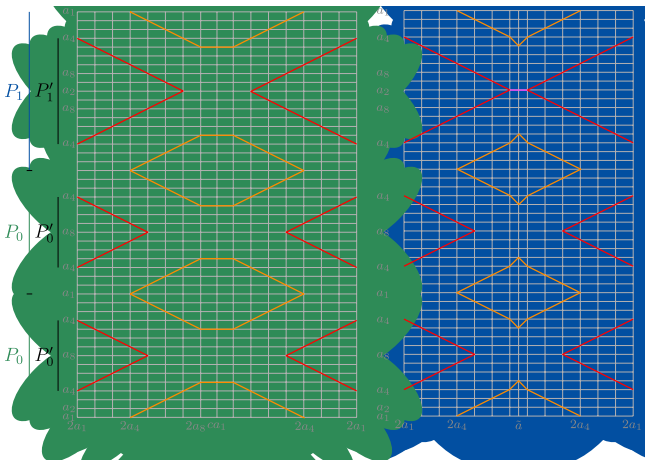


Figure 3: In white: the non-free space of the curve Q_0 and $P_0 \circ P_0 \circ P_1$ on the left and of the curve Q_1 and $P_0 \circ P_0 \circ P_1$ on the right. The red, orange and pink paths visualize the proof of Lemma 6.

Proof.

We show this lemma by constructing paths in the non-free space for $\delta = 1 + c$ (see Figure 3). First observe that $\|2a_i - a_j\| \geq 1 + c$ for any $i, j \in \{1, \ldots, 8\}$ with $i - j \geq 3 \mod 8$.

Assume that the first (resp. last) vertex of Q_y is matched to a point on a subcurve P_x' with $x \in \{0,1\}$. Then, there must exist a path

completely contained in the free space ending at the last (resp. first) vertex of Q_y. This cannot be the case since there exists a continuous matching between P'_x and $\langle 2a_1, 2a_2 \ldots, 2a_5, \ldots, 2a_1\rangle$ that is completely contained in the non-free space (see the red paths in Figure 3). Therefore, the first as well as the last vertex of Q_y has to be matched to a point on a subcurve P'' of P such that $P'' = \langle a_4, a_3, a_2, a_1, a_2, a_3, a_4\rangle$.

Next, we show that P' contains exactly one P'_x as a subcurve. Since $\|ca_1 - a_5\| \geq 1 + c$ (resp. $\|\tilde{a} - a_5\| \geq 1 + c$), there exists a path that is completely contained in the non-free space between $\langle a_5, a_4, \ldots a_2, a_1, a_2, \ldots, a_4, a_5\rangle$ and $\langle 2a_8, ca_1, 2a_8, 2a_7, \ldots, 2a_4, 2a_5, \ldots, 2a_8, ca_1, 2a_8\rangle$(resp. $\langle \tilde{a}, 2a_7, \ldots, 2a_4, 2a_5, \ldots, \tilde{a}\rangle$) (see the orange paths in Figure 3). Hence by construction of P, it follows that P'_x is a subcurve of P' and P' is a subcurve of $\langle a_4, a_3, a_2\rangle \circ P_x \circ \langle a_2, a_3, a_4\rangle$ with $x \in \{0, 1\}$. It remains to show that $x \neq 1$, if $y = 1$. This follows since the distance between a_2 and $2a_7$, $\tilde{a}$, $2a_7$ is at least $1 + c$ (see the pink path in Figure 3). ◀ The next lemma follows directly by the construction of the curves and Lemma 6.

Lemma 7.

Consider the 2D setting and let $Q' = \bigcirc_{\ell=1}^{d'} Q_{\sigma_\ell}$, where $\sigma_\ell \in \{0, 1\}$ for every ℓ. Further, let $s \leq t$ be such that $P[s, t]$ is a subcurve in between the two subcurves P_{null} and $d_{\mathrm{F}}(Q', P[s, t]) < 1 + c$. Then, $P'_{\pi_1} \circ \bigcirc_{\ell=2}^{d'-1} P_{\pi_\ell} \circ P'_{\pi_{d'}}$ is a subcurve of $P[s, t]$ and $P[s, t]$ is a subcurve of $\langle a_4, a_3, a_2\rangle \circ \bigcirc_{\ell=1}^{d'} P_{\pi_\ell} \circ \langle a_2, a_3, a_4\rangle$ where $\pi_\ell = 0$ if $\sigma_\ell = 1$ and $\pi_\ell \in \{0, 1\}$ if $\sigma_\ell = 0$ for every ℓ.

Lemma 8.

If $d_{\mathrm{dF}}(P, Q) < 2$ (resp. $d_{\mathrm{F}}(P, Q) < 1 + c$) in the 1D (resp. 2D) construction, then there exist two indices i and j such that $\langle v_i, u_j\rangle = 0$.

Proof.

Let M be a discrete (resp. continuous) matching that realizes the discrete (resp. continuous) Fréchet distance between P and Q. Assume that $d_{\mathrm{F}}(P, Q) < 2$ (resp. $d_{\mathrm{dF}}(P, Q) < 1 + c$), then in the matching M, the two points of value P_{null} of P must be matched to two distinct points q_1, q_2 on Q both on an edge that contains the value Q_{null}, since they are the only points within distance at most $1 + c \leq 3$ to P_{null}. Let $P'_0 = P_0$ and $P'_1 = P_1$ in the 1D setting. Define j such that q_1 is contained on an incident edge to the j-th Q_{null} on Q. Then, the first $(Q_1)^{d+1}$ after the j-th Q_{null} on Q must be matched to a subcurve P' of P such that $P'_1 \circ (P_0)^{d+1} \circ P'_1$ is a subcurve of P' by Lemma 5 and Lemma 7. The only time we have $(P_0)^{d+1}$ in P is before $P_1 \circ \bigcirc_{\ell=1}^{d} P_{v_{i\ell}}$ for an i. Hence by Lemma 5 and Lemma 7, there exists an index i such that in the matching M the subcurve $P'_{v_{i,1}} \circ \bigcirc_{\ell=2}^{d-1} P_{v_{i,\ell}} \circ P'_{v_{i,d}}$ is matched to a subcurve of $\bigcirc_{\ell=1}^{d} Q_{u_{j,\ell}}$ and $v_{i,\ell} = 0$ if $u_{j,\ell} = 1$. Hence, it holds that $\langle v_i, u_j\rangle = 0$. ◀ By Lemma 4 and Lemma 8 together with Observation 3, the next two theorems follow.

Theorem 9.

For any $\delta, \varepsilon > 0$, there does not exist a $(2-\varepsilon)$-approximation algorithm for the discrete Fréchet distance between 1-dimensional curves of complexity n and m using $\mathcal{O}((nm)^{1-\delta})$ time unless OVH fails.

Theorem 10.
For any $\delta, \varepsilon > 0$, there does not exist a $(k-\varepsilon)$-approximation algorithm for the continuous or discrete Fréchet distance between 2-dimensional curves of complexity n and m using $\mathcal{O}((nm)^{1-\delta})$ time unless OVH fails, where $k = 3$ (resp. $k = 1+\sqrt{2} \approx 2.41$) if the curves lie in the L_∞-space (resp. in the Euclidean space).

Our construction gives also a bound on Fréchet distance oracles (proof in the full version).

5 $(3+\varepsilon)$-approximation Algorithm for the Fréchet distance

In this section, we give a $(3+\varepsilon)$-approximation algorithm to compute the continuous/discrete Fréchet distance between two curves P and Q of complexity n resp. m with $m \leq n$. We describe this section for the continuous Fréchet distance, but everything works analogously for the discrete Fréchet distance as well. We begin with the decision variant for δ. In the first step, using the curve Q, we compute a curve P' of complexity at most $2m$ such that $d_{\mathrm{F}}(P, P') \leq \delta$ or decide that $d_{\mathrm{F}}(P, Q) > \delta$ in linear time.

Lemma 21.
Algorithm 3 computes a curve P' of complexity at most $2m$ with $d_{\mathrm{F}}(P, P') \leq \delta$ or returns $d_{\mathrm{F}}(P, Q) > \delta$ in $\mathcal{O}(n)$ time.

Proof.
In the i-th iteration of the while-loop, it holds that $s_i \geq i$ by line 3 of Algorithm 3. Further, it holds that $a_i \leq a_{i+1}$ for every i and line 6 has running time $\mathcal{O}(a_{i+1} - a_i)$ as $\overline{Q(s_i)Q(t_i)} \subset \overline{Q(s_i)Q(\lfloor s_i + 1 \rfloor)}$. Hence, the running time follows. If a curve P' is returned, it holds that $d_{\mathrm{F}}(P, P') \leq \delta$ by line 6 and Observation 2. The complexity of P' is at most $2m$ since there are at most m iterations of the while loop. It remains to show correctness for the case when $d_{\mathrm{F}}(P, Q) > \delta$ is returned. Assume that $d_{\mathrm{F}}(P, Q) > \delta$ is returned and in fact it holds $d_{\mathrm{F}}(P, Q) \leq \delta$. Let M be the matching between P and Q that realizes the Fréchet distance. We consider two cases (see Figure 6). In the first case, there exist b, b' and an index i such that $(b, s_i), (b', t_i) \in M$ with $b \leq a_i$ and $b' > a_{i+1}$. This contradicts line 6 of the algorithm. Otherwise, there must exist an index i and a parameter c such that $(a_i, c) \in M$ with $t_{i-1} \leq c < s_i$. This contradicts line 3. Hence, it follows that $d_{\mathrm{F}}(P, Q) > \delta$. ◀

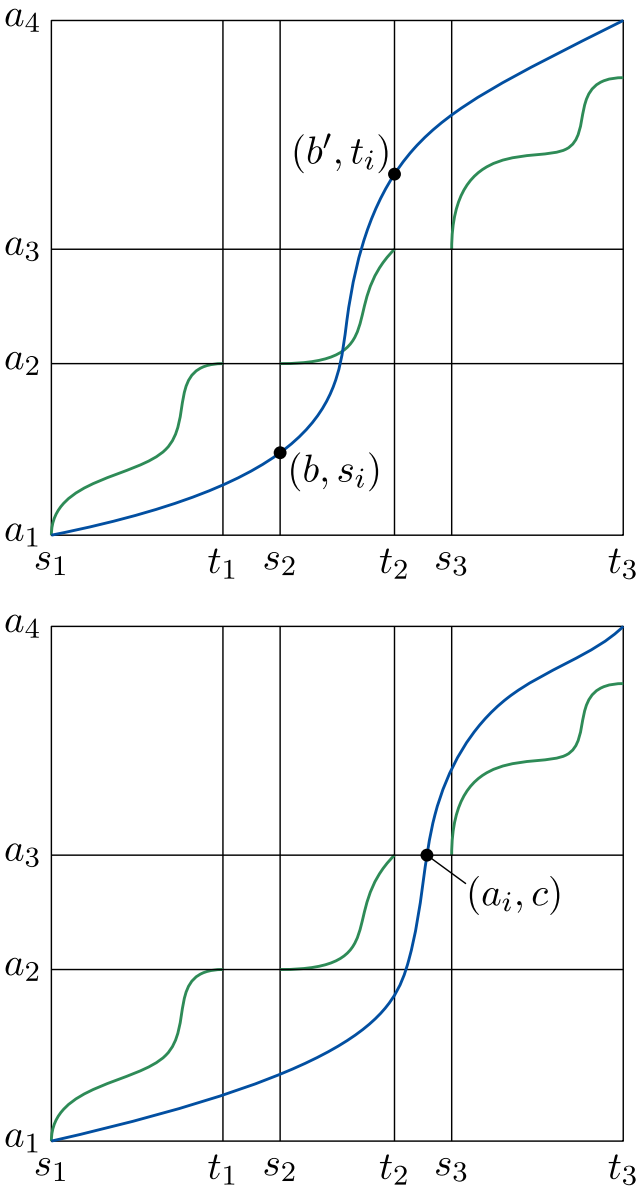


Figure 6: The green curve is a matching between the curves $P[a_i, a_{i+1}]$ and $Q[s_i, t_i]$ computed in Algorithm 3. The blue curves visualize the matching used in the proof of Lemma 21. Algorithm 3 Simplification of P depending on Q.

$s_1 \leftarrow 1$, $a_1 \leftarrow 1$, $i \leftarrow 1$
while *True* **do**
  $s_i \leftarrow \min\{s \geq \lfloor s_{i-1} + 1 \rfloor \mid \|P(a_i) - Q(s)\| \leq \delta\} \cup \{\infty\}$
  if $s_i = \infty$ **then**
    Return $d_F(P, Q) > \delta$
  Find largest a_{i+1} such that $\exists\, t_i \leq \lfloor s_i + 1 \rfloor$ with $d_F(P[a_i, a_{i+1}], \overline{Q(s_i)Q(t_i)}) \leq \delta$
  if $a_{i+1} = n$ **then**
    Return $P' = \langle Q(s_1), Q(t_1), Q(s_2), Q(t_2), \ldots, Q(s_i), Q(t_i) \rangle$
  $i \leftarrow i + 1$

Lemma 21 can be easily adapted such that it holds for the discrete Fréchet distance. Given P', we decide whether $d_{\mathrm{F}}(P', Q) \leq 2\delta$ in $\mathcal{O}(m^2)$ time [3] or slightly faster [15]. If $d_{\mathrm{F}}(P', Q) \leq 2\delta$, then $d_{\mathrm{F}}(P, Q) \leq d_{\mathrm{F}}(P', Q) + d_{\mathrm{F}}(P, P') \leq 3\delta$. Otherwise, it holds that $d_{\mathrm{F}}(P, Q) \geq d_{\mathrm{F}}(P', Q) - d_{\mathrm{F}}(P, P') > \delta$. Hence, we obtain the following lemma.

Lemma 22.

Given two curves P and Q of complexity n and m in any metric space M. Assume that we can compute distances in M in constant time. We can 3-approximate the decision variant of the continuous/discrete Fréchet distance in $\mathcal{O}(n + m^2)$ time.

For curves in the Euclidean space, Colombe and Fox (Theorem 11 of [18]) show how to transform a decision algorithm into an approximate optimization algorithm increasing the running time by a $\mathcal{O}(\log(n/\varepsilon))$ factor. Since all L_p-norms on $\mathbb{R}^d$ are equivalent and Lemma 22 holds for all L_p-norms, we can adopt their approach such that we can use it for all L_p-norms.

Theorem 23.

There exist a $(3 + \varepsilon)$-approximation algorithm for the continuous/discrete Fréchet distance using $\mathcal{O}((n + m^2) \log(n/\varepsilon))$ time.

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Author Algorithm 3, editable transcription of the official published algorithm image

$s_1 \leftarrow 1,\ a_1 \leftarrow 1,\ i \leftarrow 1$
while $True$ **do**
  $s_i \leftarrow \min\{s \geq \lfloor s_{i-1} + 1 \rfloor \mid \|P(a_i) - Q(s)\| \leq \delta\} \cup \{\infty\}$
  if $s_i = \infty$ **then**
    Return $d_{\mathrm{F}}(P, Q) > \delta$

6 Find largest a_{i+1} such that $\exists\ t_i \leq \lfloor s_i + 1 \rfloor$ with $d_{\mathrm{F}}(P[a_i, a_{i+1}], \overline{Q(s_i)Q(t_i)}) \leq \delta$
7 **if** $a_{i+1} = n$ **then**
8 Return $P' = \langle Q(s_1), Q(t_1), Q(s_2), Q(t_2), \ldots, Q(s_i), Q(t_i) \rangle$
9 $i \leftarrow i + 1$